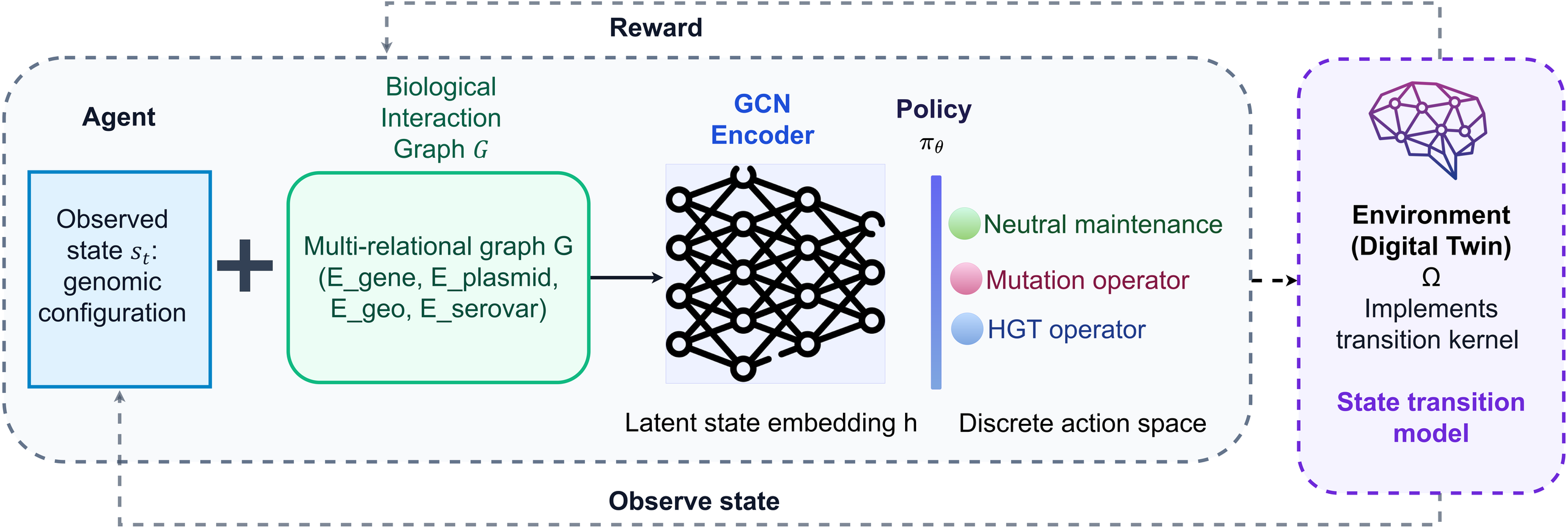
Reward
Agent
Biological Interaction Graph G
GCN Encoder
Policy
π_θ
Observed state s_t: genomic configuration
Multi-relational graph G (E_gene, E_plasmid, E_geo, E_serovar)
Neutral maintenance
Mutation operator
HGT operator
Environment (Digital Twin)
Ω
Implements transition kernel
State transition model
Latent state embedding h
Discrete action space
Observe state